LAS VIEWER

Well Log Visualization & Petrophysics Platform

DATABRICKS SOLUTION ACCELERATOR

Powered by Databricks Apps • Unity Catalog • Serverless

ENTERPRISE-GRADE WELL LOG VISUALIZATION & AI-POWERED PETROPHYSICS

LAS Viewer combines interactive SVG-based log curve visualization, automated quality control, configurable processing recipes, and Foundation Model API-powered petrophysical advisory into a single platform for upstream well log analysis.

◆ BUSINESS VALUE

- **Accelerate log interpretation** with interactive multi-track SVG display of raw, corrected, and derived curves
- **Improve data quality** with automated QC detecting spikes, gaps, range violations, and washouts
- **Standardize workflows** with configurable processing recipes from corporate standard to high-fidelity reservoir sim
- **Enhance decision-making** with AI-powered petrophysical recommendations for completion strategies
- **Centralize well data** in Lakebase PostgreSQL with formation tops, curve quality, and processing history

◆ CHALLENGES ADDRESSED

- **Scattered log data** — LAS files spread across network drives with no centralized access
- **Manual QC** — hours spent visually scanning curves for anomalies and data gaps
- **Inconsistent processing** — different petrophysicists applying different correction workflows
- **Limited AI integration** — no automated interpretation or completion recommendation capability
- **Poor visualization** — legacy desktop tools with no web-based collaborative viewing

◆ WHO IT'S FOR

Petrophysicist
Interactive log curve display, QC dashboards, and processing recipe management

Drilling Engineer
Well status overview, formation boundaries, and data quality scores

Reservoir Engineer
Derived porosity, water saturation, and clay volume for characterization

Completions Engineer
AI-powered advisor for well completion and development strategies

6	9	6	3
Wells	Log Curves	Dashboard Tabs	Processing Recipes

◆ QUICK START GUIDE

1. **Clone Repository**
   Clone the LAS Viewer repo to your local machine
2. **Build Frontend**
   Run cd frontend && npm install && npm run build
3. **Deploy the App**
   Run databricks apps deploy to launch the platform
4. **Explore Logs**
   App auto-seeds sample well data on first startup

◆ GETTING THE MOST OUT OF LAS VIEWER

Browse Log Curves
Use the interactive SVG viewer to display raw, QC-flagged, corrected, and derived curves

Run Automated QC
Check per-curve quality scores for spike detection, gap identification, and washout flagging

Apply Recipes
Execute corporate standard, fast turnaround, or high-fidelity processing workflows

Consult the Advisor
Get AI-powered petrophysical recommendations for completion strategies

◆ TARGET INDUSTRY

Upstream Oil & Gas — Well logging and petrophysical analysis for exploration, appraisal, and development wells across any basin or formation type.

⚠ **DEMO ONLY** — This application is a demonstration and solution accelerator. Please refer to the included LICENSE file for full terms of use.

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